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Homework 1: Circuit Emulation (Three Ways)

CIRCUIT DESCRIPTION

`Emulate a circuit that takes in 3 inputs (A, B, C) and whose output Q is True only when at least 2 of the 3 inputs are False.`

Introduction

I worked my way down the paper. I started with the truth table, then wrote the Boolean expression. After writing the boolean expression, I simplified it and finally completed the circuit diagram. The truth table was the easiest of the given tasks because of all the practice I've had in this class. The circuit diagram was the most challenging because it was new, and I wasn't sure if I was doing it efficiently. I struggled to avoid getting lost in the wires, often connecting the wrong ends and having to restart multiple times.

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Truth Table/Test Table

A	B	C	Q (expected)	R (actual)	Subexpression
0	0	0	1	1	~A ^ ~B ^ ~C
0	0	1	1	1	~A ^ ~B ^ C
0	1	0	1	1	~A ^ B ^ ~C
0	1	1	0	0	-
1	0	0	1	1	A ^ ~B ^ ~C
1	0	1	0	0	-
1	1	0	0	0	-
1	1	1	0	0	-

Boolean Algebra Expression

Q= (~A ^~B ^ ~C) v (~A ^ ~B ^ C) v (~A ^ B ^ ~C) v (A ^ ~B ^ ~C)

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[Optional for extra credit] Simplification

Starting boolean algebra expression
(~A ^ ~B ^ ~C) v (~A ^ ~B ^ C) v (~A ^ B ^ ~C) v (A ^ ~B ^ ~C)
Step 1. Use the commutative law & associative law to group first two terms
(~A ^~B ^ ~C) v (~A ^ ~B ^ C)
Step 2. factor out the common term using the distributive law
~A ^ ~B ^ (~C v C)
Step 3 complement law
(~C v C) = 1
Step 4 identity law
~A ^ ~B ^ 1 = ~A ^ ~B
Step 5 Associative law for regrouping
(~A ^ ~B) v (~A ^ B ^~C) v (A ^ ~B ^ ~C)
Step 6 Use the distributive law to factor out the common term
~C ^ ((~A ^ B) v (A ^ ~B))
Step 7: Bring expressions back together
(~A ^ ~B) v ~C ^ ((~A ^ B) v (A ^ ~B))

Circuit Diagram